



Afatinib versus methotrexate as second-line treatment in patients with recurrent or metastatic squamous-cell carcinoma of the head and neck progressing on or after platinum-based therapy (LUX-Head & Neck 1): an open-label, randomised phase 3 trial

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Summary

Background Patients with recurrent or metastatic squamous-cell carcinoma of the head and neck (HNSCC) progressing after first-line platinum regimens have a poor prognosis and few treatment options. Afatinib, an irreversible ERBB family blocker, has shown efficacy in a phase 2 study in this setting. We aimed to assess the efficacy and safety of afatinib compared with methotrexate as second-line treatment in patients with recurrent or metastatic HNSCC progressing on or after platinum-based therapy.

Methods In this open-label, phase 3, randomised controlled trial conducted in 101 centres in 19 countries, we enrolled patients aged 18 years or older with histologically or cytologically confirmed HNSCC that was recurrent, metastatic, or both who had progressed on or after first-line platinum-based therapy, were not amenable for salvage surgery or radiotherapy, and who had an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. Previous treatment with more than one systemic regimen in this setting was not allowed; previous treatment with EGFR-targeted antibody therapy (but not EGFR-targeted tyrosine-kinase inhibitors) was allowed. We randomly assigned eligible patients in a 2:1 ratio to receive oral afatinib (40 mg/day) or intravenous methotrexate (40 mg/m² per week), stratified by ECOG performance status and previous EGFR-targeted antibody therapy for recurrent or metastatic disease. Randomisation was done centrally with an interactive voice or web-based response system. Clinicians and patients were not masked to treatment allocation; independent review of tumour response was done in a blinded manner. The primary endpoint was progression-free survival as assessed by an independent, central imaging review committee. Efficacy analyses were done in the intention-to-treat population and safety analyses were done in patients who received at least one dose of study drug. This ongoing study is registered with ClinicalTrials.gov, number NCT01345682.

Findings Between Jan 10, 2012, and Dec 12, 2013, we enrolled 483 patients and randomly assigned 322 to afatinib and 161 to methotrexate. After a median follow-up of 6·7 months (IQR 3·1–9·0), progression-free survival was longer in the afatinib group than in the methotrexate group (median 2·6 months [95% CI 2·0–2·7] for the afatinib group vs 1·7 months [1·5–2·4] for the methotrexate group; hazard ratio [HR] 0·80 [95% CI 0·65–0·98], $p=0\cdot030$). The most frequent grade 3 or 4 drug-related adverse events were rash or acne (31 [10%] of 320 patients in the afatinib group vs none of 160 patients in the methotrexate group), diarrhoea (30 [9%] vs three [2%]), stomatitis (20 [6%] vs 13 [8%]), fatigue (18 [6%] vs five [3%]), and neutropenia (1 [$<1\%$] vs 11 [7%]); serious adverse events occurred in 44 (14%) of afatinib-treated patients and 18 (11%) of methotrexate-treated patients.

Interpretation Afatinib was associated with significant improvements in progression-free survival and had a manageable safety profile. These findings provide important new insights into the treatment of this patient population and support further investigations with irreversible ERBB family blockers in HNSCC.

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Introduction

Patients with recurrent or metastatic squamous-cell carcinoma of the head and neck (HNSCC) who progress after platinum-based therapy have a dismal prognosis and no well-defined standard treatment.^{1,2} In a retrospective analysis¹ of outcomes in patients with HNSCC who were refractory to platinum-based therapy, median survival was

roughly 1·8 months with best supportive care. Methotrexate, taxanes, and cetuximab (a monoclonal antibody targeting EGFR) are used alone or in combination (eg, with paclitaxel and cetuximab) as second-line therapy after platinum failure, despite the lack of randomised trials demonstrating improvement in outcomes or disease-related symptoms.^{3,4} Cetuximab monotherapy is approved

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See Online for appendix

Research in context

Evidence before this study

We did a systematic review of the scientific literature published up to Sept 30, 2014, using PubMed and of trials presented as abstracts at major oncology meetings (ie, annual meetings of the American Society of Clinical Oncology and the European Society for Medical Oncology). Using the search terms "phase 2" or "phase 3" and "recurrent" or "metastatic HNSCC", we reviewed articles and presentations reporting phase 2 and 3 trials investigating anticancer agents (chemotherapy or targeted therapies) in patients with recurrent or metastatic squamous-cell carcinoma of the head and neck (HNSCC) who progressed on or after platinum therapy. This review of the published literature and congress presentations showed an unmet medical need for patients in this setting, with no well-defined standard of care and few efficacious treatment options.

Added value of this study

This study demonstrates that the irreversible ERBB-family blocker afatinib has clinical efficacy in second-line recurrent or

metastatic HNSCC, with a significant improvement in progression-free survival and patient-reported outcomes compared with chemotherapy after platinum failure. Interestingly, we noted a more pronounced benefit for progression-free survival in patients with baseline characteristics potentially linked to human papillomavirus (HPV) negativity. To our knowledge, no studies, including this trial, have shown an improvement in overall survival in this difficult-to-treat population.

Implications of all the available evidence

In light of the unmet medical need, these data provide important new insights into the treatment of patients with recurrent or metastatic HNSCC progressing on or after platinum therapy and support further investigations with irreversible ERBB family blockers in HNSCC, particularly in patients with HPV-negative disease.

for second-line treatment in this population,⁵ but only in the USA; it is also often used in earlier treatment lines.^{6,7} In prospective studies of second-line methotrexate or cetuximab after platinum therapy in patients with recurrent or metastatic HNSCC, median survival ranged between 3 months and 6 months.^{2,8,9}

The ERBB receptor family network, comprising EGFR (HER1, ERBB1), HER2 (ERBB2), HER3 (ERBB3), and HER4 (ERBB4), plays an important part in tumorigenesis.¹⁰ About 80–90% of patients with HNSCC overexpress EGFR, and this overexpression negatively affects prognosis.⁴ Crosstalk among ERBB receptors could decrease the clinical efficacy of EGFR-targeted therapies and promote treatment resistance.¹¹ One strategy to improve efficacy and overcome resistance is to inhibit several ERBB receptors simultaneously.

Afatinib, an irreversible ERBB-family blocker,^{12,13} showed efficacy in a phase 2 study of patients with recurrent or metastatic HNSCC after failure of platinum-based therapy.¹⁴ Here, we report results of a phase 3 trial comparing the efficacy and safety of afatinib with that of methotrexate in patients with recurrent or metastatic HNSCC who had progressed on or after first-line platinum-based chemotherapy. We chose methotrexate as the comparator group in this study because this single-agent chemotherapy is used globally in recurrent or metastatic HNSCC^{15,16} and was a comparator for studies of other targeted agents in HNSCC.^{8,9}

Methods

Study design and participants

LUX-Head & Neck 1 is a randomised, global, phase 3, open-label trial done at 101 centres in 19 countries worldwide. Eligible patients were aged 18 years or older;

had histologically or cytologically confirmed squamous-cell carcinomas of the oral cavity, oropharynx, hypopharynx, or larynx that was recurrent, metastatic, or both and were not amenable for salvage surgery or radiotherapy; had an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1; and had a life expectancy of at least 3 months. Patients had documented evidence of progression on the basis of investigator assessment (measurable disease according to Response Evaluation Criteria in Solid Tumors [RECIST] version 1.1)¹⁷ after two or more cycles of cisplatin (≥ 60 mg/m² per cycle or total ≥ 120 mg/m² over 8 weeks) or carboplatin (area under the concentration-time curve ≥ 4 per cycle or total ≥ 8 over 8 weeks) for recurrent or metastatic disease; previous treatment with more than one systemic regimen in this setting was not allowed. Previous treatment with EGFR-targeted antibody therapy (but not EGFR-targeted tyrosine-kinase inhibitors) was allowed.

Additional key exclusion criteria included progressive disease within 3 months of completing curatively intended treatment; primary tumour site of nasopharynx (of any histology), sinuses, or salivary glands; treatment with any investigational drug or anticancer therapy within 4 weeks before randomisation (except palliative radiotherapy to bones to alleviate pain); unresolved chronic toxic effects (other than grade >2 hearing loss, tinnitus, or dry mouth from previous anticancer therapy or unresolved grade >1 skin toxicities or diarrhoea caused by previous treatment with EGFR-targeted antibodies); previous grade 3 or higher tumour bleeding; major surgical or planned procedure less than 4 weeks before randomisation (we did not classify isolated biopsies as major surgical procedures); any other malignant disease (except for HNSCC of a location described in the inclusion criteria, basal-cell carcinoma, or

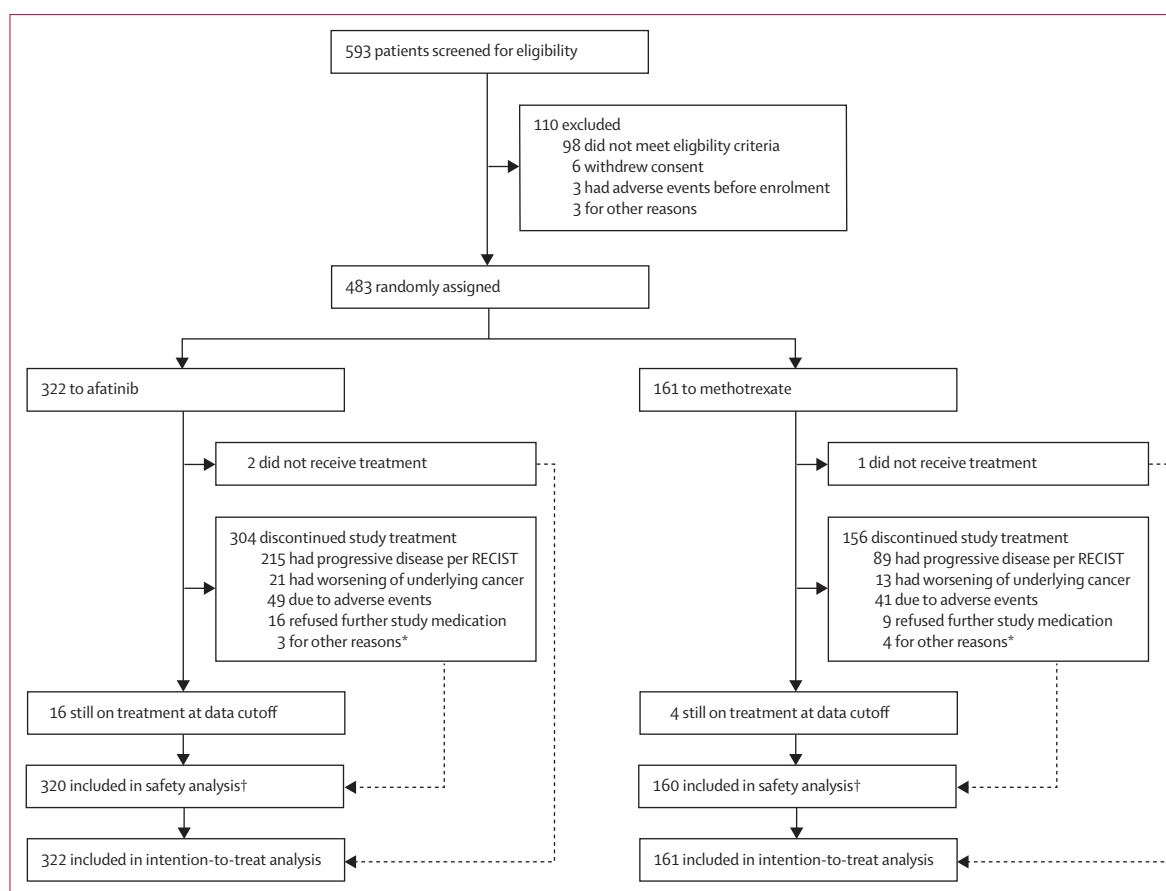


Figure 1: Study design

RECIST=Response Evaluation Criteria In Solid Tumors. *Includes "patient or investigator decision", "non-compliant with protocol", and "lost to follow-up". †Safety analyses included randomly assigned patients who received at least one dose of study drug.

surgically cured cervical cancer in situ); known brain metastasis or signs thereof; known pre-existing interstitial lung disease; clinically relevant cardiovascular abnormalities (eg, uncontrolled hypertension, congestive heart failure with a New York Heart Association classification of 3, unstable angina, or poorly controlled arrhythmia) or myocardial infarction within 6 months before randomisation; cardiac left ventricular function with resting ejection fraction of less than 50%; active hepatitis B or C infection or known HIV carrier; clinically significant or recent acute gastrointestinal disorders with diarrhoea as a major symptom; and laboratory values of absolute neutrophil count less than 1500 cells per μL , platelet count less than 75 000 cells per μL , creatinine clearance less than 50 mL/min, bilirubin more than 1.5 times upper limit of normal (ULN), and aspartate aminotransferase or alanine aminotransferase more than three times ULN (if related to liver metastases, more than five times ULN).

The study protocol, designed in accordance with the Declaration of Helsinki, the International Conference on Harmonization Guideline for Good Clinical Practice, and applicable region-specific regulatory requirements, was approved by independent ethics committees at each centre.

All patients provided written informed consent for trial participation. Patients who volunteered for assessment of tumour biomarkers provided informed consent for collection and analysis of tissue samples.

Randomisation and masking

We randomly assigned patients in a 2:1 ratio to receive afatinib or methotrexate, stratified by ECOG performance status (0 vs 1) and previous use of EGFR-targeted antibody therapy for recurrent or metastatic disease (yes vs no). Randomisation was done with a validated random-number generating system at Boehringer Ingelheim, verified by a trial-independent statistician, and implemented centrally with an interactive voice or web-based response system.¹⁸ Individuals directly involved in the conduct and analysis of the trial did not have access to the randomisation schedule. Due to the different administration routes of the study drugs, patients and investigators were not masked to treatment allocation. To minimise bias, all patients participated in weekly visits, and the primary analysis was based on assessments by an independent, central imaging review committee who evaluated tumour scans or images for each patient in a blinded manner.

	Afatinib (n=322)	Methotrexate (n=161)
Sex		
Men	275 (85%)	137 (85%)
Women	47 (15%)	24 (15%)
Age, years		
Median (range)	60 (32–82)	59 (32–88)
<65	239 (74%)	116 (72%)
≥65	83 (26%)	45 (28%)
ECOG performance status		
0	89 (28%)	42 (26%)
1	233 (72%)	119 (74%)
Smoking pack-years		
<10	56 (17%)	31 (19%)
≥10	255 (79%)	126 (78%)
Unknown	11 (3%)	4 (2%)
Alcohol consumption, units per week		
≤7	250 (78%)	124 (77%)
>7	58 (18%)	33 (20%)
Unknown	14 (4%)	4 (2%)
Primary tumour site		
Oral cavity	94 (29%)	42 (26%)
Oropharynx	99 (31%)	54 (34%)
Hypopharynx	63 (20%)	30 (19%)
Larynx	66 (20%)	35 (22%)
Time since first diagnosis, years		
Median (range)	2.0 (0.3–27.1)	2.2 (0.5–21.7)
<2	163 (51%)	71 (44%)
≥2	158 (49%)	89 (55%)
Unknown	1 (<1%)	1 (<1%)
Localisation of recurrence		
Locoregional disease only	106 (33%)	61 (38%)
Distant metastases only	46 (14%)	18 (11%)
Both	164 (51%)	77 (48%)
Unknown	6 (2%)	5 (3%)
Metastatic sites*		
Non-regional lymph nodes	80 (25%)	39 (24%)
Lung	155 (48%)	76 (47%)
Liver	37 (11%)	15 (9%)
Bone	32 (10%)	13 (8%)
Skin	3 (<1%)	0
Other	34 (11%)	13 (8%)
Unknown	3 (<1%)	1 (<1%)
p16 status†		
Positive	31 (10%)	18 (11%)
Negative	141 (44%)	67 (42%)
Not assessed	150 (47%)	76 (47%)
Differentiation grade		
Well differentiated	62 (19%)	31 (19%)
Moderately differentiated	112 (35%)	55 (34%)
Poorly differentiated	93 (29%)	46 (29%)
Undifferentiated	3 (<1%)	3 (2%)
Not specified or unknown	52 (16%)	26 (16%)

(Table 1 continues on next page)

Procedures

Once-daily oral afatinib was given at an initial dose of 40 mg and increased to 50 mg after at least 4 weeks of treatment if drug-related adverse events were minimal. In case of drug-related grade 3 or selected grade 1–2 adverse events, we reduced the afatinib dose in 10 mg decrements to a minimum of 20 mg. The starting dose of methotrexate was 40 mg/m² (intravenous bolus injection, once weekly), and could be increased to 50 mg/m² after at least 2 weeks of treatment and no evidence of drug-related adverse events. In case of drug-related adverse events, we reduced the methotrexate dose in 10 mg/m² decrements to a minimum of 20 mg/m². Treatment continued until disease progression, unacceptable toxic effects, or other reasons necessitating withdrawal. After disease progression, if a patient was judged by the investigator as deriving clinical benefit (eg, improvement in disease-related symptoms) from the study drug despite progression, the patient was allowed to continue his or her randomised treatment.

We performed tumour assessments with CT or MRI, and response was assessed by investigators and independent central review (according to RECIST version 1.1) every 6 weeks for the first 24 weeks, and every 8 weeks thereafter. Patient-reported outcomes were assessed with the European Organization for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire (QLQ-C30)¹⁹ and the head and neck cancer-specific supplementary module (QLQ-H&N35).²⁰

We monitored concomitant drugs (any non-oncological therapy, except for vitamins or appetite stimulants) and safety weekly, with incidence and intensity of adverse events graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 3.0. Safety laboratory assessments (urinalysis, biochemistry, and haematology) were performed after 1 week of treatment, every 3 weeks for the first 24 weeks, and every 4 weeks thereafter. Details about data collection after disease progression are included in the appendix.

Tumour biomarker assessment of p16 status was conducted in a central or local laboratory (as described in the appendix) with use of archival tumour tissue samples from all patients who voluntarily provided separate informed consent for the analysis.

Outcomes

The primary endpoint was progression-free survival, defined as the time from randomisation to the time of progression or death, whichever occurred first. The primary analysis of progression-free survival was based on assessment by an independent, central imaging review committee. Overall survival, defined as the time from randomisation to the time of death, was the key secondary endpoint; other endpoints included objective response (defined as complete response or partial response according to RECIST version 1.1), tumour shrinkage (defined as the maximum decrease from baseline in the sum of diameters of the target lesions), patient-reported outcomes related to disease symptoms and

quality of life (assessed by EORTC QLQ-C30 and QLQ-H&N35), and safety. Prespecified analyses of patient-reported outcomes included time to deterioration (greater than or equal to ten-point worsening from baseline); distribution of patients whose symptoms had improved (greater than or equal to ten-point improvement from baseline), remained stable, or worsened; and mean difference in symptom scores over time. These analyses focused on the following scales: pain (composite of items 31–34 of the EORTC QLQ-H&N35), swallowing (composite of items 35–38 of the EORTC QLQ-H&N35), and global health status and quality of life (composite of items 29 and 30 of the EORTC QLQ-C30). We also assessed disease control (ie, objective response or stable disease). Prespecified subgroup analyses included baseline ECOG score (0 vs 1), previous use of EGFR-targeted antibody for recurrent or metastatic HNSCC (yes vs no), sex (men vs women), age (<65 years vs ≥65 years), region (Asia vs Europe vs North America and Latin America), smoking history (<10 pack-years vs ≥10 pack-years), alcohol consumption (seven or fewer units per week vs greater than seven units per week), primary tumour site (oral cavity vs oropharynx vs hypopharynx vs larynx), recurrence or metastases (recurrent vs metastatic vs both), p16 status (positive vs negative), and response to previous platinum therapy for recurrent or metastatic HNSCC (complete response, partial response, or stable disease vs progressive disease).

Statistical analysis

The study was powered to demonstrate a benefit of afatinib versus methotrexate in progression-free survival and overall survival, with a planned randomisation of 474 patients. 364 progression-free survival events were needed to achieve 90% power with a one-sided type I error (α) of 0.025, assuming a median progression-free survival of 3.0 months with afatinib and 2.1 months with methotrexate.^{8,14} We based the assumptions for median progression-free survival on results previously reported in a phase 2 study of afatinib¹⁴ and a phase 3 study of methotrexate⁸ in patients with recurrent or metastatic HNSCC. 343 deaths were needed for 80% power with a one-sided type I error (α) of 0.025, assuming median overall survival of 8.9 months with afatinib on the basis of findings from a phase 2 study comparing afatinib with cetuximab,¹⁴ and 6.5 months with methotrexate.⁹

Efficacy analyses included all randomised patients (the intention-to-treat population). Safety analyses included all treated patients (randomised patients receiving at least one dose of study drug). We analysed the primary and key secondary endpoints following a hierarchical testing procedure: stratified log-rank test with strata used for randomisation on progression-free survival first (afatinib vs methotrexate), followed by overall survival. Statistical tests reported two-sided p values. We used the Kaplan-Meier method to estimate progression-free survival and overall survival for each treatment group, and derived hazard ratios (HRs) for afatinib versus methotrexate using

	Afatinib (n=322)	Methotrexate (n=161)
(Continued from previous page)		
Previous platinum-based therapy for recurrent or metastatic disease		
Cisplatin	168 (52%)	83 (52%)
Carboplatin	119 (37%)	47 (29%)
Cisplatin and carboplatin	30 (9%)	28 (17%)
Other	5 (2%)	3 (2%)
Previous use of anti-EGFR monoclonal antibody for recurrent or metastatic disease†	189 (59%)	98 (61%)
Duration of anti-EGFR monoclonal antibody for recurrent or metastatic disease, weeks‡		
≤12	48 (15%)	19 (12%)
>12 to 24	70 (22%)	31 (19%)
>24	69 (21%)	49 (30%)
Previous surgery	244 (76%)	122 (76%)
Previous curative anticancer therapy		
Radiotherapy only	77 (24%)	37 (23%)
Chemoradiation therapy	145 (45%)	73 (45%)
Previous curative chemotherapy		
Neoadjuvant or induction	64 (20%)	37 (23%)
Adjuvant¶	35 (11%)	17 (11%)

Data are n (%) or median (range). ECOG=Eastern Cooperative Oncology Group. *Total numbers of patients with metastatic sites were 211 in the afatinib group and 98 in the methotrexate group. †Based on either local or central test results. ‡One patient in the afatinib group received previous panitumumab and the remaining patients received cetuximab. §Duration of anti-EGFR monoclonal antibody therapy for recurrent or metastatic disease was unknown for two patients in the afatinib group; in the methotrexate group, one patient who reported use of anti-EGFR monoclonal antibody therapy on the case report form was determined to have not received previous anti-EGFR monoclonal antibody therapy. ¶24 patients in the afatinib group and eight patients in the methotrexate group received previous curative adjuvant chemotherapy in combination with radiotherapy after surgery, and are thus reported for both chemoradiation therapy and adjuvant chemotherapy.

Table 1: Patient demographics and baseline characteristics

a stratified Cox proportional hazards model. For predefined subgroup analyses, we used a Cox proportional hazards model to explore subgroup by treatment interactions. Progression-free survival by investigator review was a predefined sensitivity analysis. We analysed subgroup variables individually. We performed an exploratory analysis of overall survival using the inverse probability of censoring weighted model,²¹ on the basis of receipt of subsequent anticancer therapies after discontinuation of study drugs. A description of how the patient-reported outcomes were analysed is in the appendix. SAS version 9.2 was used for all statistical analyses. This study is registered with ClinicalTrials.gov, number NCT01345682.

Role of the funding source

The trial was designed by the LUX-Head & Neck 1 steering committee in collaboration with experts in the specialty and the funder. Data were collected by the study investigators. An independent data monitoring committee monitored study conduct. The funder's study team, in collaboration with the study investigators, was responsible for the analysis and interpretation of the data; all authors had access to the data. The funder coordinated the development of the report, which was written by the corresponding author in collaboration with the coauthors,

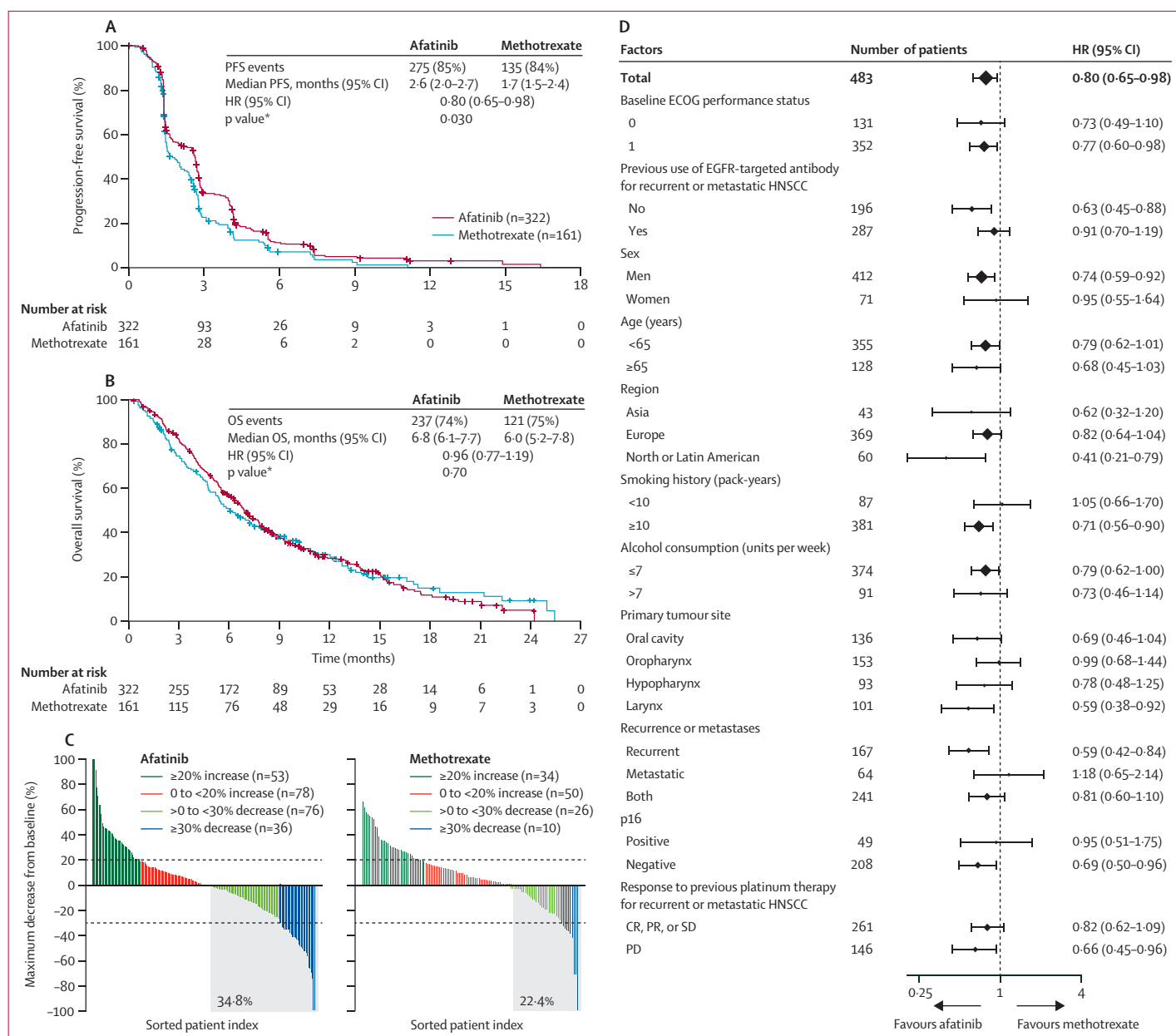


Figure 2: Survival outcomes, tumour shrinkage, and subgroup analysis of progression-free survival

(A) Kaplan-Meier estimates of progression-free survival. (B) Kaplan-Meier estimates of overall survival. (C) Waterfall plot of maximum percentage tumour shrinkage. Dotted lines show the ≥20% increase and ≥30% decrease cutoffs. (D) Forest plot of progression-free survival according to predefined subgroups. PFS=progression-free survival. HR=hazard ratio. OS=overall survival. ECOG=Eastern Cooperative Oncology Group. HNSCC=squamous-cell carcinoma of the head and neck. CR=complete response. PR=partial response. SD=stable disease. PD=progressive disease. *Stratified log-rank test.

with independent medical writing assistance supported financially by the funder. All authors were responsible for the report's content, were involved at all stages of its development, and approved the final version. The corresponding author had responsibility for the final decision to submit for publication.

Results

We screened 593 patients for eligibility, and between Jan 10, 2012, and Dec 12, 2013, we enrolled 483 patients

and randomly assigned them to receive afatinib (n=322) or methotrexate (n=161); reasons for patient ineligibility for randomisation are detailed in figure 1. Two patients in the afatinib group (one did not meet eligibility criteria and one withdrew consent after randomisation) and one in the methotrexate group (decided not to take the study drug) did not receive treatment. All randomised patients were included in the efficacy analyses; 480 patients were included in the safety population. Table 1 shows the baseline characteristics of the treatment groups.

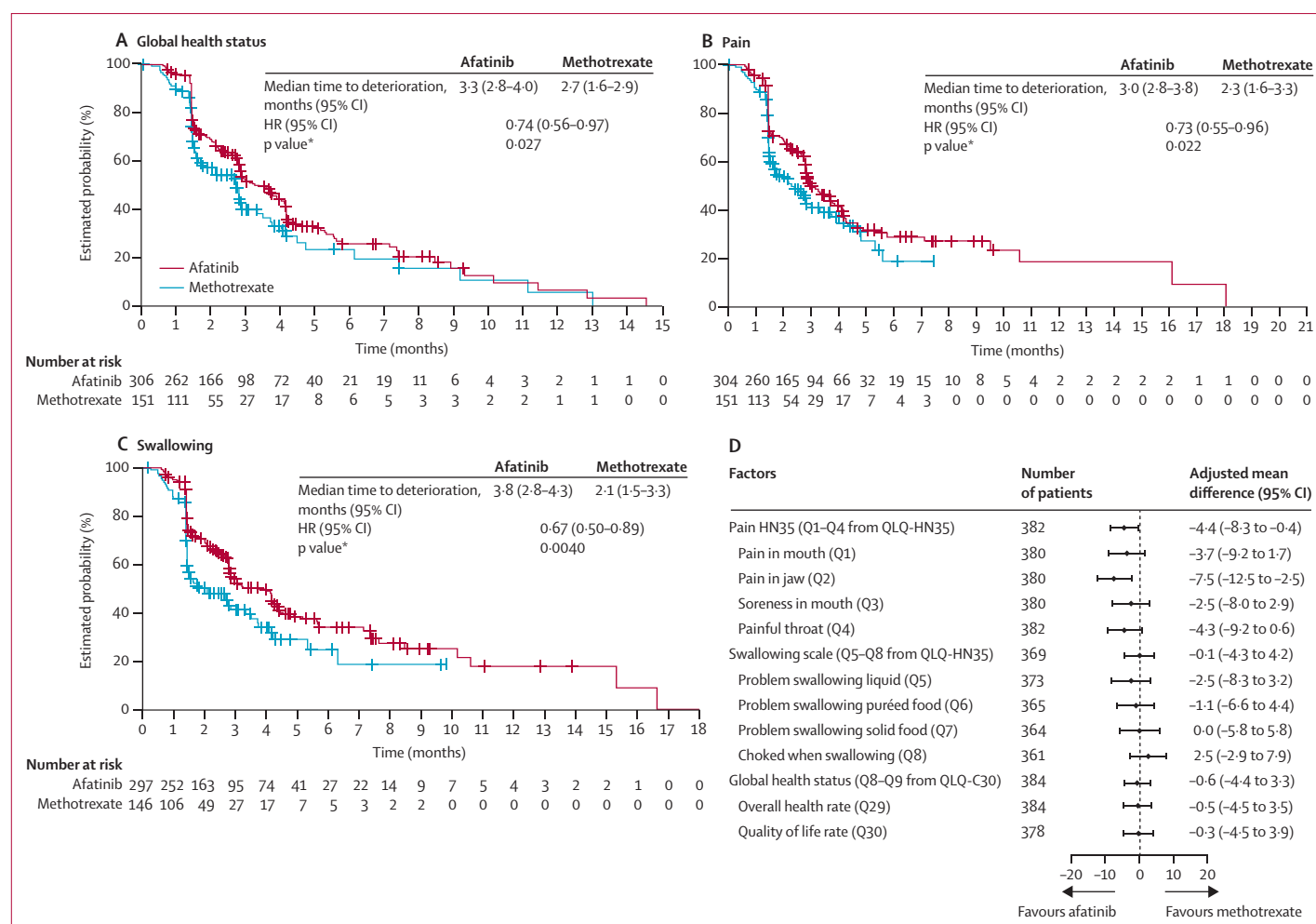


Figure 3: Patient-reported quality-of-life outcomes

(A) Kaplan-Meier estimates of deterioration over time in EORTC global health status. (B) Kaplan-Meier estimates of deterioration over time in EORTC pain. (C) Kaplan-Meier estimates of deterioration over time in EORTC swallowing. (D) Longitudinal analysis of differences in mean scores over time. Q=question. EORTC QLQ-C30=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire. HN35=EORTC QLQ head and neck cancer-specific supplementary module. HR=hazard ratio. *Stratified log-rank test.

The data cutoff for analysis of progression-free survival was May 7, 2014, and for overall survival was June 30, 2014, after a median follow-up of 6.7 months (IQR 3.1–9.0). Median treatment duration was 83 days (42–130) with afatinib and 43 days (29–85) with methotrexate. Most patients took at least 80% of the assigned drug (271 [89%] of 306 patients for afatinib and 135 [84%] of 160 patients for methotrexate; 14 patients in the afatinib group did not have data available for calculation of compliance).

At the time of this analysis, 275 (85%) of 322 patients in the afatinib group and 135 (84%) of 161 patients in the methotrexate group experienced a progression-free survival event (figure 2A). Median progression-free survival by independent review was 2.6 months (95% CI 2.0–2.7) with afatinib versus 1.7 months (1.5–2.4) with methotrexate (HR 0.80 [95% CI 0.65–0.98], stratified log-rank test $p=0.030$; figure 2A). By investigator review,

median progression-free survival was 2.7 months (95% CI 2.6–2.8) and 1.9 months (1.5–2.5) with afatinib and methotrexate, respectively (HR 0.79 [95% CI 0.64–0.98], $p=0.026$). At the time of this analysis, 237 (74%) of 322 patients in the afatinib group and 121 (75%) of 161 patients in the methotrexate group had died (figure 2B). Median overall survival was 6.8 months (95% CI 6.1–7.7) with afatinib and 6.0 months (5.2–7.8) with methotrexate (HR 0.96 [95% CI 0.77–1.19], $p=0.70$; figure 2B).

An objective response (all partial responses), assessed by independent review, occurred in 33 (10%) of 322 patients given afatinib and nine (6%) of 161 patients given methotrexate ($p=0.10$); disease control was achieved in 158 (49%) patients given afatinib (125 [39%] patients had stable disease) compared with 62 (39%) patients given methotrexate (53 [33%] patients had stable disease; $p=0.035$; appendix). Tumour shrinkage from baseline according to independent central review was observed in

	Afatinib (n=320)				Methotrexate (n=160)			
	Grade 1-2	Grade 3	Grade 4	Grade 5	Grade 1-2	Grade 3	Grade 4	Grade 5
Any adverse event	176 (55%)	114 (36%)	11 (3%)	2 (<1%)	80 (50%)	44 (28%)	8 (5%)	5 (3%)
Rash or acne*	207 (65%)	31 (10%)	0	0	13 (8%)	0	0	0
Diarrhoea	201 (63%)	28 (9%)	2 (<1%)	0	16 (10%)	3 (2%)	0	0
Stomatitis†	105 (33%)	19 (6%)	1 (<1%)	0	56 (35%)	13 (8%)	0	0
Fatigue‡	61 (19%)	18 (6%)	0	0	46 (29%)	5 (3%)	0	0
Nausea	59 (18%)	5 (2%)	0	0	35 (22%)	1 (<1%)	0	0
Paronychia§	43 (13%)	3 (<1%)	0	0	0	0	0	0
Vomiting	37 (12%)	3 (<1%)	1 (<1%)	0	14 (9%)	0	0	0
Dry skin	36 (11%)	0	0	0	0	0	0	0
Decreased appetite	33 (10%)	10 (3%)	0	0	20 (13%)	1 (<1%)	0	0
Weight decreased	24 (8%)	2 (<1%)	0	0	5 (3%)	0	0	0
Pruritus	22 (7%)	4 (1%)	0	0	0	0	0	0
Anaemia	18 (6%)	4 (1%)	0	0	20 (13%)	8 (5%)	2 (1%)	0
Conjunctivitis¶	16 (5%)	2 (<1%)	0	0	2 (1%)	0	0	0
Palmar-plantar erythrodysesthesia	15 (5%)	2 (<1%)	0	0	2 (1%)	0	0	0
Abdominal pain	11 (3%)	2 (<1%)	0	0	0	0	0	0
Hypomagnesaemia	7 (2%)	1 (<1%)	0	0	2 (1%)	0	0	0
Xerosis	7 (2%)	1 (<1%)	0	0	0	0	0	0
Hypokalaemia	6 (2%)	3 (<1%)	4 (1%)	0	3 (2%)	1 (<1%)	0	0
Dyspnoea	4 (1%)	1 (<1%)	0	0	2 (1%)	1 (<1%)	0	0
Glossitis	4 (1%)	1 (<1%)	0	0	0	0	0	0
Aspartate aminotransferase increased	4 (1%)	0	0	0	11 (7%)	3 (2%)	1 (<1%)	0
Dysphagia	3 (<1%)	3 (<1%)	0	0	0	0	0	0
Gastro-oesophageal reflux disease	3 (<1%)	0	1 (<1%)	0	0	0	0	0
Lymphopenia	3 (<1%)	0	0	0	1 (<1%)	1 (<1%)	2 (1%)	0
Dehydration	2 (<1%)	5 (2%)	0	0	0	0	0	0
Hyponatraemia	2 (<1%)	2 (<1%)	2 (<1%)	0	0	1 (<1%)	1 (<1%)	0
Interstitial lung disease	2 (<1%)	1 (<1%)	0	0	1 (<1%)	1 (<1%)	0	0
Oesophagitis	2 (<1%)	1 (<1%)	0	0	1 (<1%)	0	0	0
Pneumonia	2 (<1%)	1 (<1%)	0	0	0	0	0	0
γ glutamyl transferase increased	2 (<1%)	0	0	0	2 (1%)	0	1 (<1%)	0
Tumour haemorrhage**	1 (<1%)	2 (<1%)	0	0	4 (3%)	0	0	0
Oral candidiasis	1 (<1%)	1 (<1%)	0	0	5 (3%)	0	0	0
Haemoptysis	1 (<1%)	1 (<1%)	0	0	2 (1%)	0	0	0
Alanine aminotransferase increased	1 (<1%)	0	0	0	12 (8%)	2 (1%)	1 (<1%)	0
Leucopenia	1 (<1%)	0	0	0	5 (3%)	8 (5%)	0	0
Platelet count decreased	1 (<1%)	0	0	0	4 (3%)	2 (1%)	1 (<1%)	0
Infection	1 (<1%)	0	0	0	1 (<1%)	1 (<1%)	0	0
Pneumonitis	1 (<1%)	0	0	0	1 (<1%)	1 (<1%)	0	0
Somnolence	1 (<1%)	0	0	0	0	1 (<1%)	0	0
Renal failure	0	2 (<1%)	0	0	0	0	0	1 (<1%)
Tracheo-oesophageal fistula	0	2 (<1%)	0	0	0	0	0	0
Renal failure acute	0	1 (<1%)	1 (<1%)	0	0	0	0	0
Neutropenia	0	1 (<1%)	0	0	20 (13%)	9 (6%)	2 (1%)	0
Gastritis	0	1 (<1%)	0	0	1 (<1%)	0	0	0
General physical health deterioration	0	1 (<1%)	0	0	0	0	0	1 (<1%)
Syncope	0	1 (<1%)	0	0	0	0	0	0
Ageusia	0	1 (<1%)	0	0	0	0	0	0
Blood creatine phosphokinase increased	0	1 (<1%)	0	0	0	0	0	0
Fungal skin infection	0	1 (<1%)	0	0	0	0	0	0
International normalised ratio	0	1 (<1%)	0	0	0	0	0	0

(Table 2 continues on next page)

	Afinib (n=320)				Methotrexate (n=160)			
	Grade 1-2	Grade 3	Grade 4	Grade 5	Grade 1-2	Grade 3	Grade 4	Grade 5
(Continued from previous page)								
Organising pneumonia	0	1 (<1%)	0	0	0	0	0	0
Pneumonia aspiration	0	1 (<1%)	0	0	0	0	0	0
Tracheobronchitis	0	1 (<1%)	0	0	0	0	0	0
Thrombocytopenia	0	0	1 (<1%)	0	4 (3%)	4 (3%)	0	0
Hypotension	0	0	1 (<1%)	0	2 (1%)	1 (<1%)	0	0
Oesophageal stenosis	0	0	1 (<1%)	0	0	0	0	0
Febrile bone marrow aplasia	0	0	1 (<1%)	0	0	0	0	0
Aspiration	0	0	0	1 (<1%)	0	0	0	1 (<1%)
Septic shock	0	0	0	1 (<1%)	0	0	0	1 (<1%)
Neutrophil count decreased	0	0	0	0	5 (3%)	1 (<1%)	0	0
White blood cell count decreased	0	0	0	0	2 (1%)	1 (<1%)	1 (<1%)	0
Hepatotoxicity	0	0	0	0	1 (<1%)	2 (1%)	0	0
Pancytopenia	0	0	0	0	0	2 (1%)	0	1 (<1%)
Hepatic function abnormal	0	0	0	0	0	1 (<1%)	0	0
Hyperbilirubinaemia	0	0	0	0	0	1 (<1%)	0	0
Hypercalcaemia	0	0	0	0	0	1 (<1%)	0	0
Mucosal haemorrhage	0	0	0	0	0	1 (<1%)	0	0
Pleural effusion	0	0	0	0	0	1 (<1%)	0	0
Staphylococcal infection	0	0	0	0	0	1 (<1%)	0	0
Stridor	0	0	0	0	0	1 (<1%)	0	0
Febrile neutropenia	0	0	0	0	0	0	2 (1%)	0
Sepsis	0	0	0	0	0	0	0	1 (<1%)

*Grouped term including the following reported preferred terms: acne, dermatitis, dermatitis acneiform, erythema, exfoliative rash, folliculitis, rash, rash erythematous, rash macular, rash maculopapular, rash pruritic, rash pustular, skin exfoliation, skin fissures, skin lesion, skin reaction, skin toxicity, and skin ulcer. †Grouped term including the following reported preferred terms: aphthous stomatitis, mucosal erosion, mucosal inflammation, mouth ulceration, and stomatitis. ‡Grouped term including the following reported preferred terms: asthenia, chronic fatigue syndrome, fatigue, and malaise. §Grouped term including the following reported preferred terms: nail bed infection and paronychia. ¶Grouped term including the reported preferred term of conjunctivitis. ||Grouped term including interstitial lung disease, organising pneumonia, and pneumonitis. **Grouped term including mouth, tracheal, and tumour haemorrhage.

Table 2: Drug-related adverse events (≥10% of patients in either treatment group with grade 1-2 and all grade ≥3 adverse events).

112 (35%) patients treated with afatinib and 36 (22%) patients treated with methotrexate (figure 2C).

In preplanned analyses of progression-free survival (independent central review) by stratification groups (figure 2D), we noted improvement with afatinib for patients who were not previously treated with an EGFR-targeted antibody for recurrent or metastatic disease. Men, patients with cancer of the larynx, patients with locally recurrent versus metastatic disease, and heavy smokers (≥10 pack-years) also seemed to derive more benefit with afatinib than with methotrexate in preplanned subgroup analyses (figure 2D). 285 (59%) of 483 patients provided tumour samples for the assessment of p16 status (234 [48%] at a central laboratory and 51 [11%] at local laboratories). High concordance was found between results in patients tested at both central and local laboratories (appendix). On the basis of these post-hoc analyses, we noted that the benefit of afatinib compared with methotrexate was more pronounced in p16-negative disease (HR 0.69 [95% CI 0.50–0.96]) than in p16-positive disease (0.95 [CI 0.51–1.75]; interaction test $p=0.32$; appendix) on the basis of central and local testing; we noted similar results for central laboratory testing only (data not shown). The proportion of patients who responded in the

different prespecified subgroups was consistent with the progression-free survival results; patients without previous treatment with EGFR-targeted therapy, with p16-negative disease, with locally recurrent disease, and with non-opharyngeal cancer had a greater response with afatinib (appendix). For both progression-free survival and tumour response analyses, no significant interactions were detected between the different subgroups. Consistent with the overall population, we did not record differences in overall survival between afatinib and methotrexate in the subgroup analyses (data not shown).

Among patients in the intention-to-treat population, 460 (95%) of 483 completed EORTC questionnaires at baseline, and participation remained high during the first 24 weeks of treatment (appendix). In prespecified analyses of patient-reported outcomes, afatinib was significantly associated with delayed time to deterioration of global health status, pain, and swallowing compared with methotrexate (figure 3A–C). Afatinib-treated patients had less pain, particularly jaw pain, over time ($p=0.030$; figure 3D), but we did not note any significant differences over time in global health status ($p=0.78$) or swallowing ($p=0.98$). The proportions of patients experiencing quality-of-life improvements with afatinib versus methotrexate (of

patients who completed questionnaires) were, respectively, 70 (26%) of 265 patients versus 27 (23%) of 117 patients for pain (odds ratio [OR] 1.19 [95% CI 0.72–1.99], $p=0.49$), 67 (26%) of 257 patients versus 26 (23%) of 112 patients for swallowing (1.16 [0.69–1.95], $p=0.58$), and 81 (30%) of 267 patients versus 34 (29%) of 117 patients for global health status (1.06 [0.66–1.71], $p=0.82$).

215 (67%) patients in the afatinib group had grade 3 or higher adverse events (127 [40%] were drug related) compared with 101 (63%) patients in the methotrexate group (57 [36%] were drug related; table 2). The most frequent grade 3 or 4 drug-related adverse events were rash or acne and diarrhoea with afatinib, and stomatitis and neutropenia with methotrexate (table 2). Few patients in the afatinib and methotrexate groups had drug-related adverse events related to interstitial lung disease (three [$<1\%$] and two [1%], respectively) or tumour haemorrhage (three [$<1\%$] and four [3%], respectively). Two (1%) patients had febrile neutropenia in the methotrexate group. Drug-related serious adverse events occurred in 44 (14%) patients treated with afatinib and 18 (11%) patients treated with methotrexate, with drug-related deaths occurring in two ($<1\%$) patients in the afatinib group (one septic shock, one aspiration pneumonia) and five (3%) patients in the methotrexate group (one septic shock, one sepsis, one aspiration pneumonia, one general health deterioration, and one renal failure and pancytopenia).

Drug-related adverse events led to dose reductions in 103 (32%) patients in the afatinib group and 67 (42%) patients in the methotrexate group, and discontinuations in 23 (7%) patients receiving afatinib and 26 (16%) patients receiving methotrexate. The most common reasons for dose reduction with afatinib were diarrhoea (34 [11%]) and rash or acne (32 [10%]); three ($<1\%$) patients discontinued treatment because of drug-related diarrhoea and none discontinued because of rash or acne. The most common reasons for dose reduction with methotrexate were stomatitis (20 [13%]) and neutropenia (21 [13%]); three (2%) and four (3%) patients, respectively, discontinued treatment due to these adverse events. As specified in the protocol, afatinib was escalated to 50 mg/day after 28 days in 74 (23%) patients who had minimal drug-related adverse events; methotrexate was escalated to 50 mg/m² in 28 (18%) patients. Dose escalation was not associated with higher rates of drug-related adverse events (data not shown).

Among the patients who discontinued study treatment, 156 (51%) of 304 patients assigned to afatinib and 79 (51%) of 156 patients assigned to methotrexate received anticancer therapy after discontinuation of study drugs (appendix). Among these patients, 25 (32%) patients in the methotrexate group received anti-EGFR therapy alone or in combination with chemotherapy (most received cetuximab; one patient received subsequent afatinib); 69 (44%) patients in the afatinib group received methotrexate. Taxanes were used as subsequent therapy in 73 (47%) patients in the afatinib group and 50 (63%) patients in the methotrexate group. In an exploratory post-hoc analysis

using the inverse probability of censoring weighted model²¹ based on receipt of subsequent anticancer therapy, median overall survival was 6.9 months (95% CI 5.9–9.8) with afatinib and 5.4 months (4.7–8.1) with methotrexate (HR 0.75 [95% CI 0.54–1.03]; appendix).

Discussion

In this randomised, open-label phase 3 trial, the primary endpoint of progression-free survival was significantly improved by afatinib compared with methotrexate in patients with second-line recurrent or metastatic HNSCC. Subgroup analyses showed progression-free survival benefit across most subgroups. We noted a pronounced benefit for progression-free survival in patients who had not previously been treated with an EGFR-targeted antibody in the recurrent or metastatic setting.

In post-hoc analyses, we found that subgroups of patients that had baseline characteristics potentially linked to human papillomavirus (HPV) negativity (ie, p16-negative disease, larynx primary site, and smoking history ≥ 10 pack-years) exhibited a greater benefit in progression-free survival with afatinib.²² The hypothesis that HPV-related HNSCC is not responsive to EGFR inhibitors, especially small molecules, has not been directly tested but is supported by recent molecular phenotyping showing EGFR-independent signalling in HPV-related HNSCC.^{23,24} Similar to other studies (NCT01302834, NCT01084083), and for practicality in this large, multicentre clinical trial, we used p16 immunohistochemistry as a surrogate for HPV infection. The reliability of p16 as a surrogate biomarker for HPV infection is based on the close correlation between p16 overexpression and HPV infection mainly in oropharyngeal cancers.^{25–27} p16 expression has been shown to have prognostic value even exclusive of HPV status.²⁸

Afatinib did not significantly improve overall survival compared with methotrexate. Median overall survival in the methotrexate group was consistent with findings from previous trials.^{8,9} The lack of an observed overall survival difference with afatinib might result from some characteristics of the study population as highlighted in the prespecified subgroup analyses, particularly the inclusion of p16-positive patients and those who received previous treatment with EGFR-targeted antibody therapy. Further, receipt of subsequent therapies after disease progression might have affected overall survival, because half the patients in each treatment group received subsequent therapy, including EGFR-targeted therapy in the methotrexate group and methotrexate in the afatinib group. Of note, the use of subsequent anticancer therapies after progression in this study was appreciably higher than in previously reported trials (14–38%).^{8,9}

We chose progression-free survival as the primary endpoint in this trial because, unlike overall survival, it is not confounded by subsequent therapies.²⁹ However, in the absence of an overall survival benefit, the modest improvement in progression-free survival noted in the

overall study population raises the question of the clinical relevance of the findings of this trial. An important issue in patients with relapsing HNSCC is the severe symptomatic burden associated with progressive disease.³⁰ In this trial, the patient-reported outcomes—measured on predefined scales—of time to deterioration of global health status, pain, and swallowing, and changes in mean scores for pain over time, were significantly improved with afatinib versus methotrexate. Further, the time to deterioration curves of global health status, pain, and swallowing were representative of the progression-free survival curves observed with each treatment, suggesting an association between prolonged control of quality of life and symptoms, and progression-free survival with afatinib. Previous phase 3 trials of EGFR-targeted drugs in the recurrent or metastatic setting have not shown significant improvements in patient-reported outcomes of global health status compared with chemotherapy,^{8,9} although improvements in disease-related symptoms of pain and swallowing were reported in patients treated with first-line cetuximab plus chemotherapy versus chemotherapy alone.³¹ Although the significant improvements in progression-free survival and patient-reported outcomes with afatinib were clinically modest, to our knowledge this is the first time that such improvements have been shown in a phase 3 trial in this setting. In this context, the clinical relevance of these findings should be further investigated and validated in additional studies of selected populations.

The treatment durations achieved with afatinib and methotrexate in the current study are consistent with those reported in the phase 2 trial of afatinib versus cetuximab in patients with recurrent or metastatic HNSCC¹⁴ (Seiwert TY, unpublished data) and the phase 3 trial of gefitinib versus methotrexate in patients with recurrent HNSCC.⁹ Of note, the current study is purely of second-line therapy in patients with recurrent or metastatic HNSCC, whereas the previously reported studies include mixed first-line and second-line treatment populations, and the option of cross-over to comparator therapy in the phase 2 study, which might have affected the duration of study treatments. The safety profile was as expected for each drug, and favoured afatinib compared with methotrexate in terms of fewer treatment-related dose reductions, discontinuations, and deaths. The most frequent drug-related adverse events with afatinib were rash or acne and diarrhoea; most were grade 1 or 2, leading to dose reductions in 11% or less of patients and few discontinuations, suggesting that these adverse events can be proactively managed and that the dose-reduction plan for afatinib in this study was effective. In this context, management of the adverse events associated with afatinib is known to be important for achievement of maximum clinical benefit.^{32,33} Although poor adherence to oral therapy is an important concern in patients with cancer,³⁴ we noted high compliance (89% of patients) with afatinib.

This study has some potential limitations. First, the study was done in an unselected population, including patients

who had received previous treatment with EGFR-targeted antibody therapy and irrespective of p16 status. As suggested by the subgroup analyses of progression-free survival, the inclusion of these patients might have diluted the treatment effect of afatinib. However, in the case of evaluation of novel agents without well-defined biomarkers in a new indication, unselected populations should be included in primary analyses to obtain a complete picture of the drug's clinical activity. Of note, patients who received previous EGFR-targeted antibody therapy were included in the study on the basis of the increasing use of cetuximab in the first-line setting. Another potential limitation is the choice of methotrexate as the comparator treatment. Although single-agent methotrexate is used globally in this setting, other drugs (eg, taxanes) might be more frequently used in some regions. Finally, in this second-line treatment population, overall survival might have been a more relevant primary endpoint than progression-free survival. However, as suggested by the 51% of patients in this study receiving subsequent therapies, second-line treatment for recurrent or metastatic HNSCC is not necessarily the last line of therapy, and the improvements in progression-free survival and patient-reported outcomes with afatinib suggest that some patients might derive substantial clinical benefit with this drug.

In summary, this trial and a previously reported randomised trial of afatinib versus cetuximab¹⁴ have shown efficacy with afatinib in patients with recurrent or metastatic HNSCC. Our findings show progression-free survival benefit associated with improvement in patient-reported outcomes with second-line afatinib versus intravenous chemotherapy, although there was no difference in overall survival. Taken together, these results support further investigations of afatinib in patients with HNSCC, particularly those with HPV-negative disease.

Contributors

J-PHM, RIH, JBV, XJC, EE, and EEWC contributed towards study conception and design. J-PHM, RIH, JF, LFL, MT, PMC, TG, DC, JJG, JG, FC, GdC, LdSV, UK, JMdC, and XJC contributed towards data collection. J-PHM, RIH, LFL, MT, JBV, PMC, TG, JJG, LdSV, UK, JMdC, EE, and EEWC contributed towards data analysis and interpretation. All authors contributed towards writing and revision of the manuscript. J-PHM, JG, FC, and EEWC contributed towards the literature search. J-PHM contributed towards the supervision of the study. LFL contributed towards administrative, technical or material support. J-PHM, RIH, JF, LFL, MT, PMC, TG, DC, JJG, JG, FC, GdC, LdSV, UK, and JMdC contributed towards patient recruitment.

Declaration of interests

J-PHM has held an advisory role with Boehringer Ingelheim without financial compensation. RIH has received clinical trial support to his institution in the form of grants from Boehringer Ingelheim, BMS, Merck, and Celgene; has held an unpaid consulting role with Boehringer Ingelheim; has received personal fees for advisory board participation from Eisai, BMS, Merck, and Celgene; and is a Head and Neck committee member for NCCN. LFL has held consulting or advisory board roles for Eisai, BMS, Amgen, Boehringer Ingelheim, Merck Serono, Debiopharm, VentiRx, AstraZeneca, SOBI, Novartis, and Bayer; received research funds to her institution from Eisai, Exelixis, Amgen, Boehringer Ingelheim, Merck Serono, Pfizer, AstraZeneca, Novartis, and Roche; received payment for travel coverages for medical meetings from Merck – Serono, Debiopharm, SOBI, and Bayer. MT has received grants and personal fees

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